Optimizing AI & ML

EcoTern Summer Bootcamp 2026
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About me

3rd-year PhD in CS at NCSU, in Prof. Xipeng Shen's **P**rogramming Systems and **I**ntelligent **C**omputing for Fu**ture** group

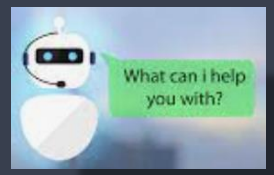
*Making computing more **intelligent** and **efficient** through innovations in system software*

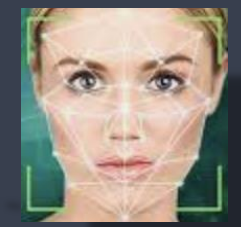
- Compilers & Code Optimizations
- High Performance Computing
- Efficient AI & Machine Learning

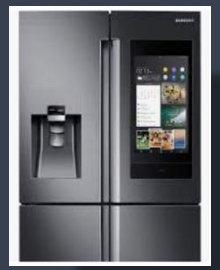
Real-time or **real time** describes various operations in computing or other processes that must guarantee response times within a specified time (deadline), usually a relatively short time.

Real-time AI describes AI operations that must guarantee response times within a specified time (deadline), usually a relatively short time.

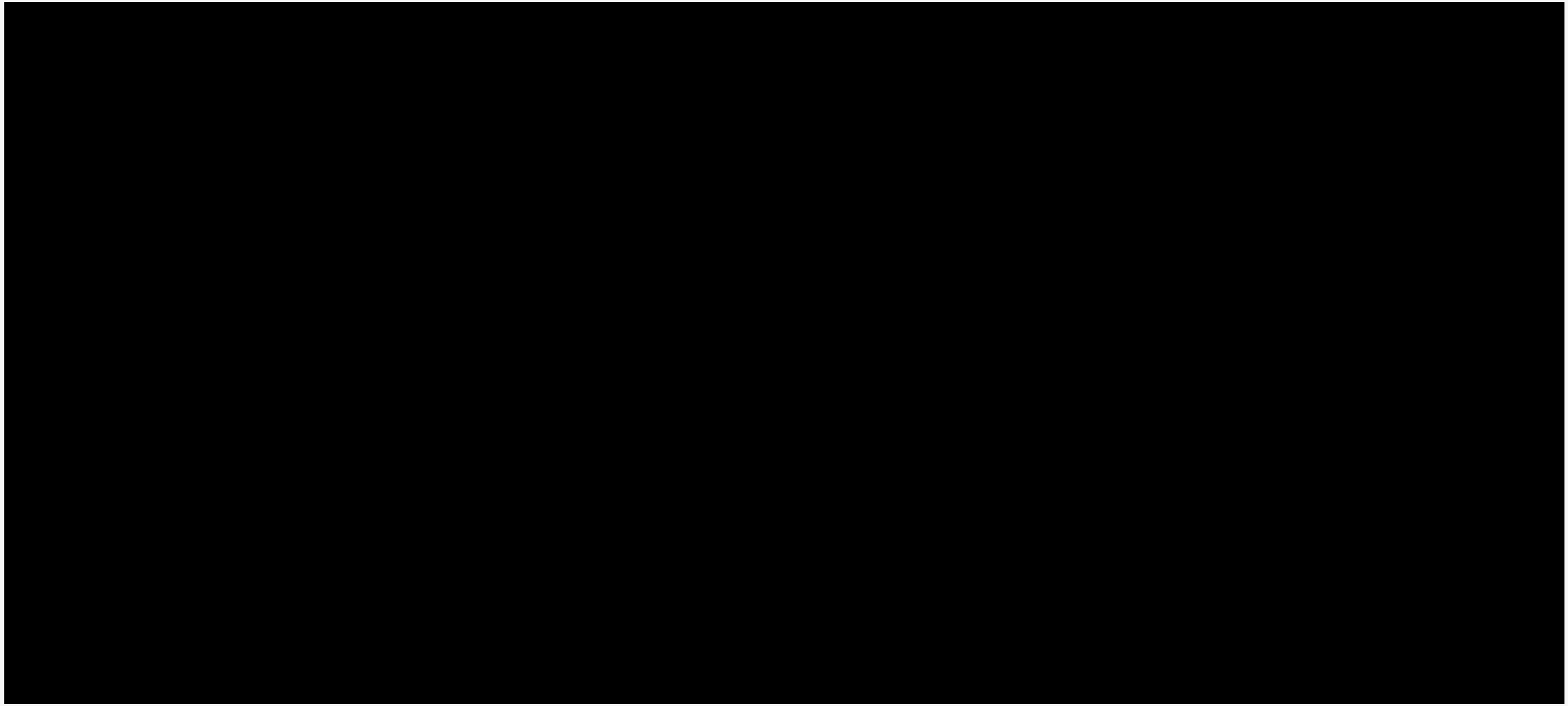
Where real-time AI is needed







Example: Realtime Super Resolution (by CoCoPIE Inc.)



High-performance computing (**HPC**)
uses supercomputers or computer clusters to solve advanced computation problems that are usually computation-intensive.

High-performance Machine Learning
uses supercomputers or computer clusters to solve machine learning problems that are computation-intensive.

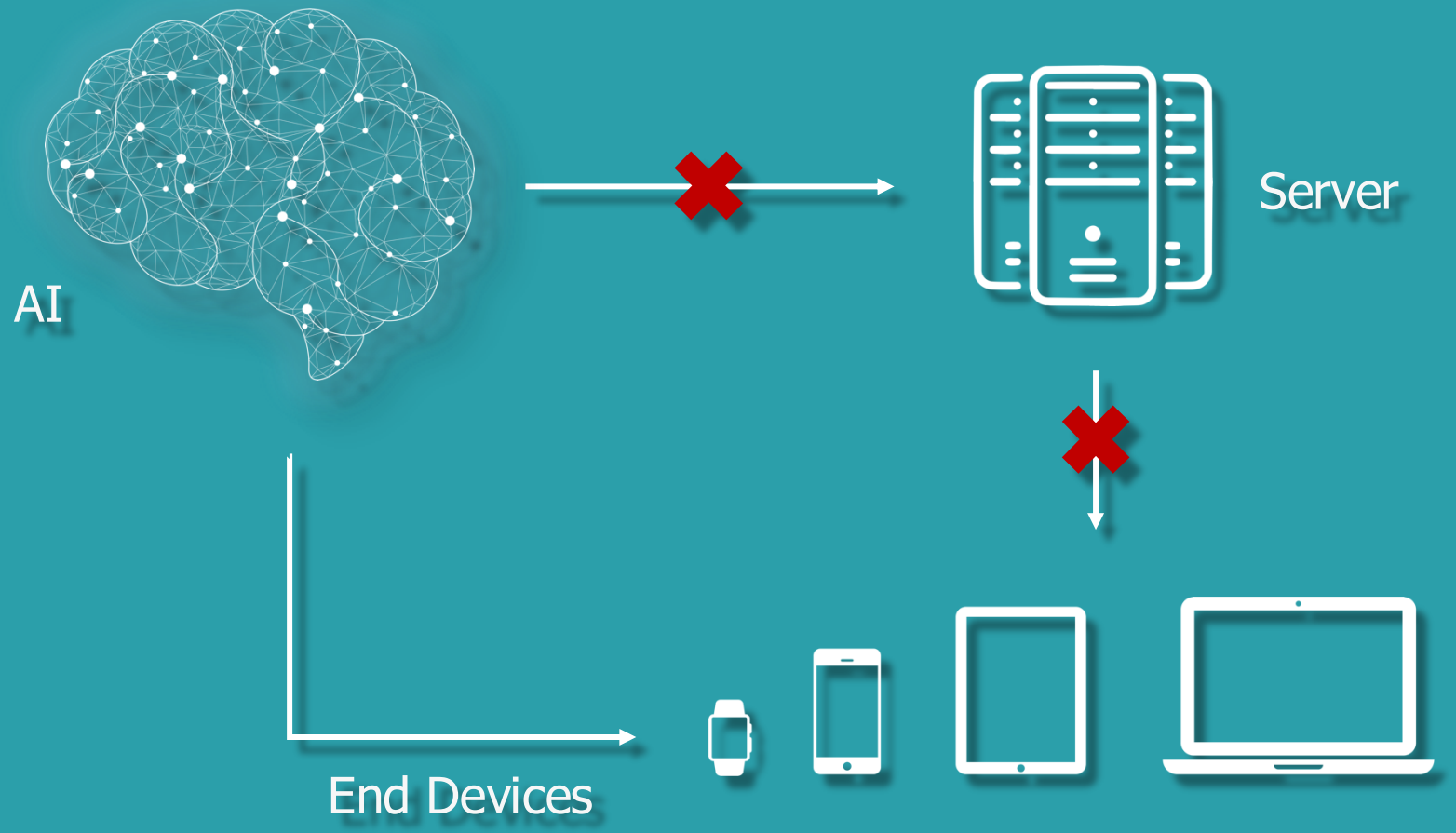
175 billion parameters

36 years to train on 8 V100 GPUs

1.7 Trillion parameters

Why is this topic important?

AI Trend: From Cloud to Device



AI Trend: From Cloud to Device

Privacy

Source: *The Verge*

AI Trend: From Cloud to Device

Real-Time

AI Trend: From Cloud to Device

Cost

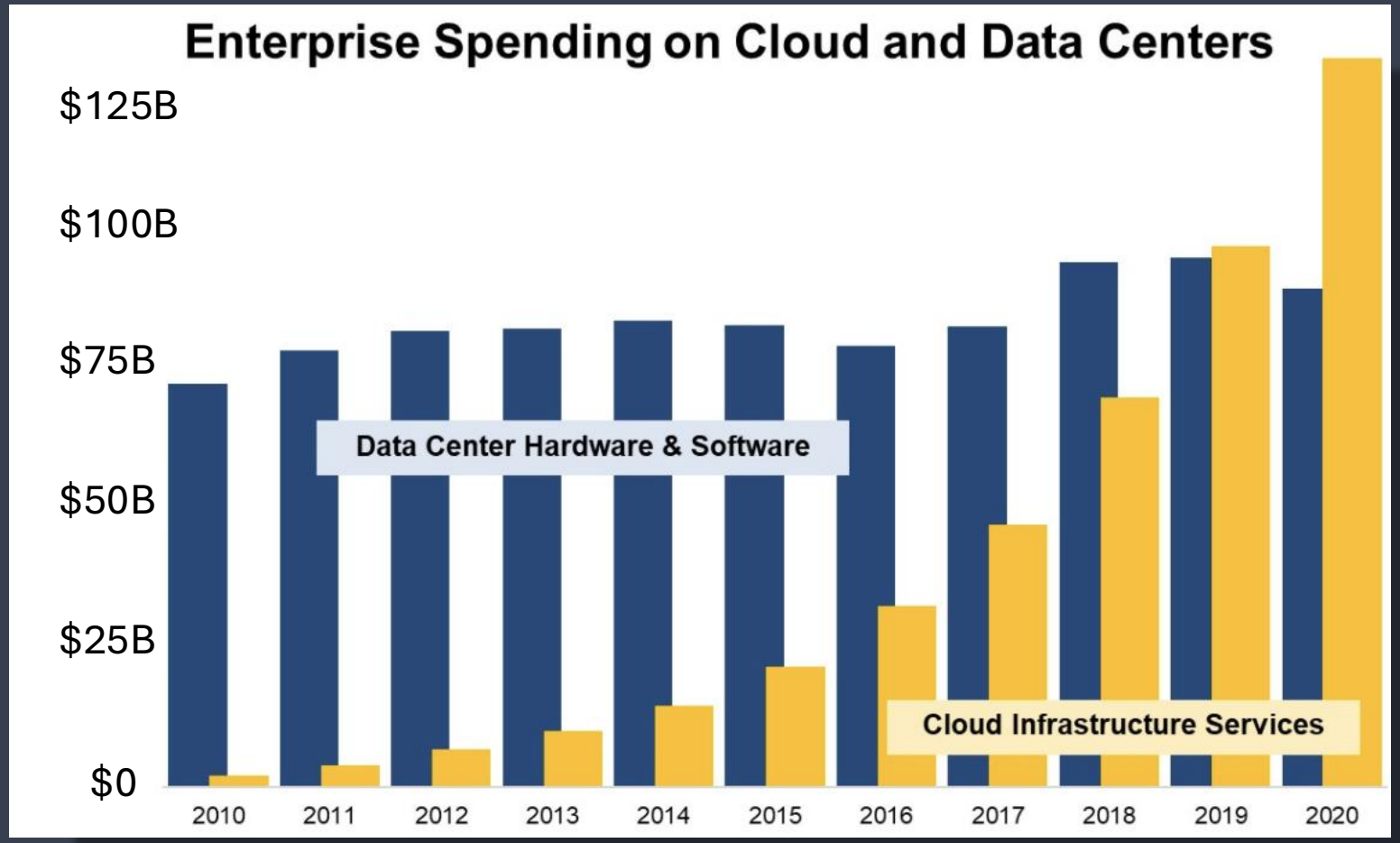
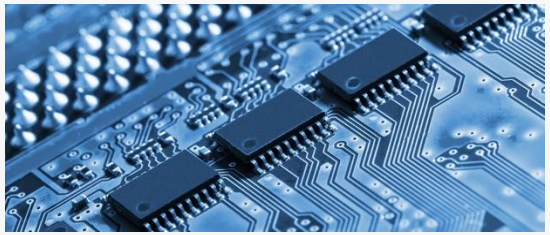


Image Credits: *Synergy Research Group*

Billions Market

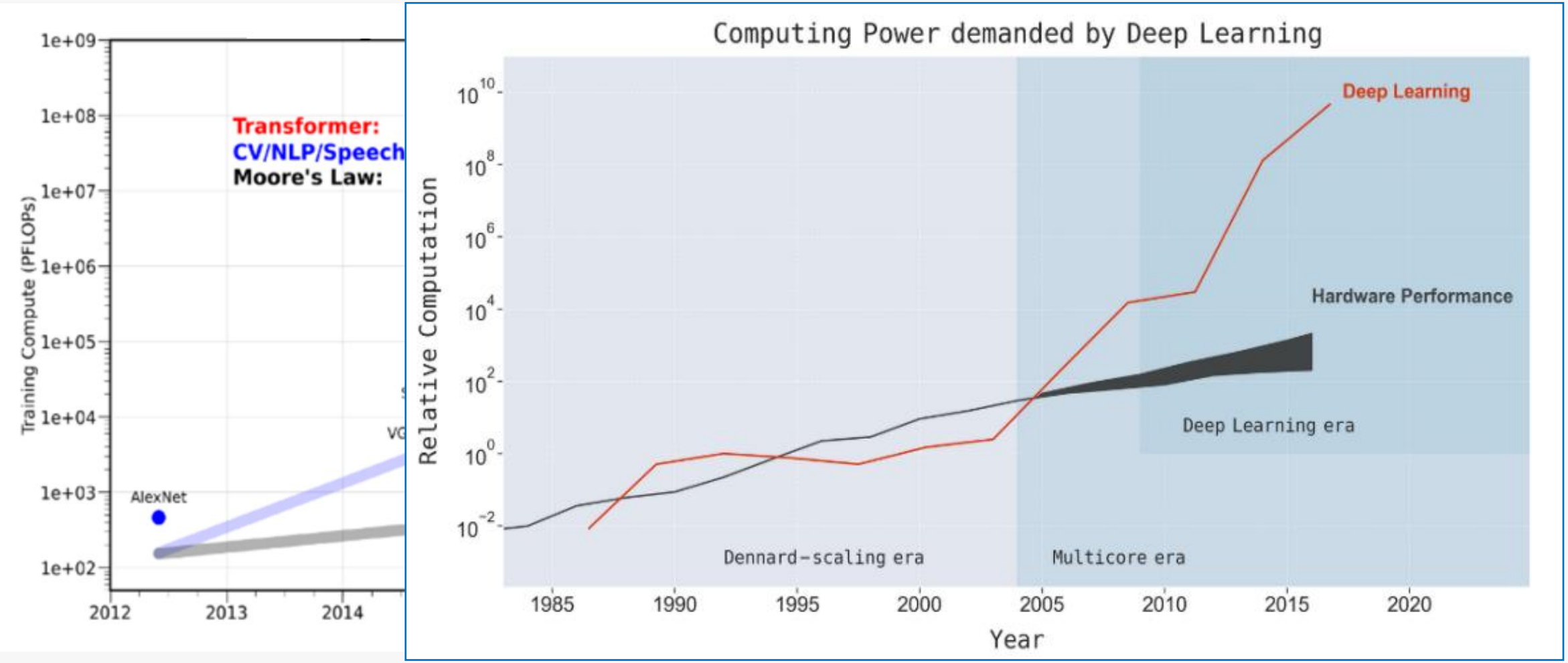
Service providers		• Media processing and enhancement • AR/VR, NLP, etc. • Personalization and recommendation • On-device training
Device makers		• Smart wearables, smart phone • Automobile, drone, robots • smart home, smart city • Smart manufacturing and farming
Chip makers		• MCU, CPU, GPU, DSP, etc. • Chips for camera, connectivity, storage, sensors, etc. • AI and acceleration chips

Consumer Electronics

Intelligent Embedded

Industry and Farms

Expanding Gap



Source: MIT INITIATIVE ON THE DIGITAL ECONOMY RESEARCH BRIEF 2020 Vol. 4

What is the current status?

Main stages

Data & Servers

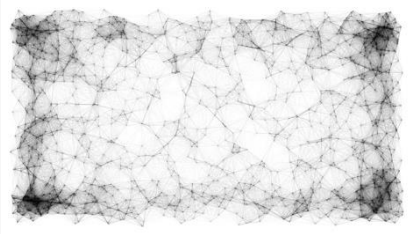
Training phase
(server-side training)

Development phase
(off-device optimization)

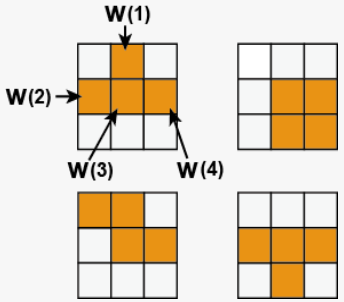
Deployment phase
(On-device execution)

ML Systems

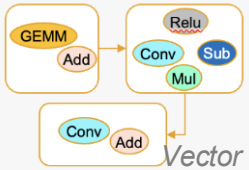
1 Model Training



2 Model Optimization



3 Code Optimizations



Vector code generation:
For (i=0; i<25; i++)
Calculate multiplication of vectors:
<a[4i]; a[4i+1]; a[4i+2]; a[4i+3]>
**<b[4i]; b[4i+1]; b[4i+2]; b[4i+3]>*

4 Runtime Optimization

5 Hardware Improvement

Active research in every stage.

Stage 1: Model Training

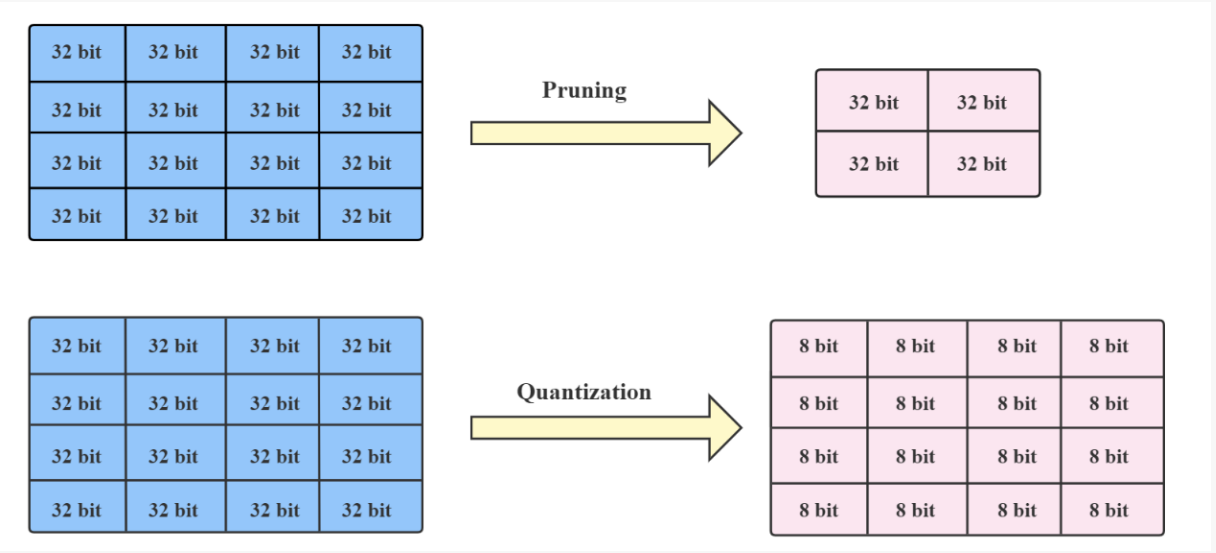
How to train faster?
How to achieve the highest accuracy?
How to deal with memory constraints of hardware?

- Distributed training frameworks
- Optimizers for backward propagation
- Memory optimizations
- Hyper parameter tuning

Stage 2: Model Optimization

How to reduce the size of a model while keeping it accurate?

- Model pruning
- Quantization
- Neural Architecture Search
- Distillation



Stage 3: Code Optimization

How to generate efficient code for a DNN?

- AI Compilation
  - Computation graph optimization
  - Parallelization & vectorization
  - Loop optimization
  - Data layout transformation
  - Kernel fusion
- Reuse of computation results

Stage 3: Code Optimization

API Fusion: in computer-vision pipelines, separate API calls (e.g., detection → segmentation → caption) repeat redundant feature extraction. Fusing them gives ~X× speed-up without accuracy loss. *(insert 1 motivation figure from your paper — e.g., the "separate vs. fused" diagram)*

Stage 4: Runtime Optimization

How to run a DNN efficiently?

- Heterogeneous scheduling
- Runtime adaptation

Stage 5: Hardware Improvement

How to make hardware best support DNNs?

- Neural Processing Units
  - Special architecture designs
- DNN-oriented features on GPUs
  - Special instructions
  - Special support of certain Tensor patterns